

Credit Risk Scorecard

Out-of-Time Validation Through the 2008 Crisis

Trained on 2000-2005 loans

Tested on 2006-2007 loans

Fannie Mae Single-Family Loan Performance Data

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What this report covers

- Builds a loan-level credit risk scorecard on 2000 to 2005 mortgage originations using predictor binning, Weight of Evidence, logistic regression, and points scaling
- Validates the scorecard out-of-time on 2006 to 2007 originations, which seasoned through the housing crisis
- Weight of Evidence (WOE) converts each predictor onto a common risk scale by comparing, within each range of values, the share of loans that repaid against the share that defaulted. Information Value (IV) sums that separation across all ranges into a single number measuring how much a predictor is worth
- Discrimination is measured by the Kolmogorov-Smirnov statistic (KS), the widest gap between defaults captured and non-defaults captured, and by the ROC curve and the area beneath it (AUC)
- Calibration asks whether the predicted level of default is right, comparing predicted against actual by score band and by origination year. A model can rank loans correctly and still badly understate how many will default
- Population stability asks whether the applicants being scored still resemble those the model was built on, measured by the Population Stability Index (PSI), the standard early warning in model monitoring
- Data prepared by the ETL pipeline: 1,178,508,035 monthly records across 48 quarterly files, reconciled to Fannie Mae published statistical summaries

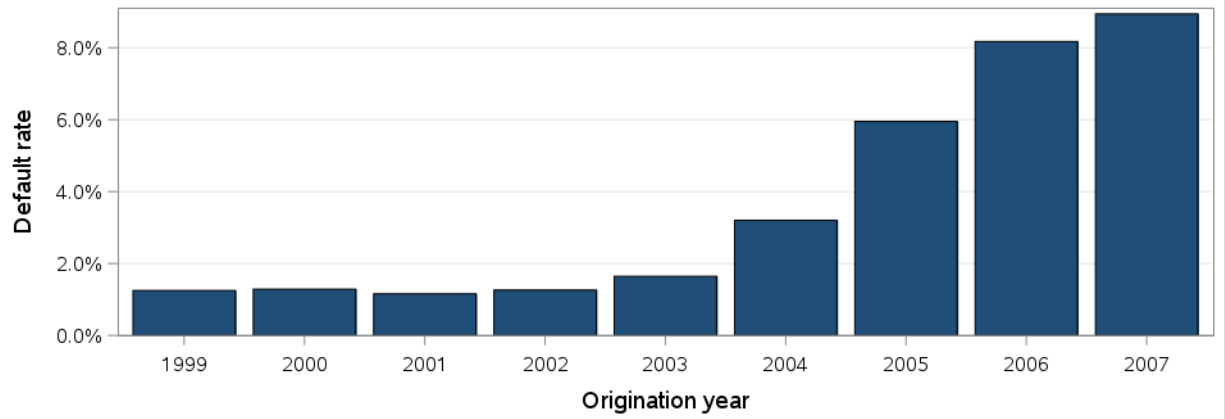
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- 19.1 million prime, fully documented, fixed-rate mortgages acquired by Fannie Mae between 2000 and 2007
- Loan counts and default rates reconcile to Fannie Mae published summaries within a few basis points on every vintage
- Default rates hold between 1.2 and 1.6 percent for loans made from 2000 through 2003
- They then climb steeply: 3.2 percent in 2004, 6.0 percent in 2005, and 8 to 9 percent in 2006 and 2007
- That is a sevenfold deterioration inside the safest segment of the mortgage market
- It happened with no comparable movement in the characteristics lenders were underwriting on

Origination year	Loans	Defaults	Default rate
1999	160,134	2,002	1.25%
2000	1,268,236	16,328	1.29%
2001	3,371,970	39,034	1.16%
2002	3,857,240	48,708	1.26%
2003	5,106,961	83,985	1.64%
2004	1,744,087	55,837	3.20%
2005	1,445,454	86,028	5.95%
2006	1,079,344	88,217	8.17%
2007	1,111,307	99,369	8.94%

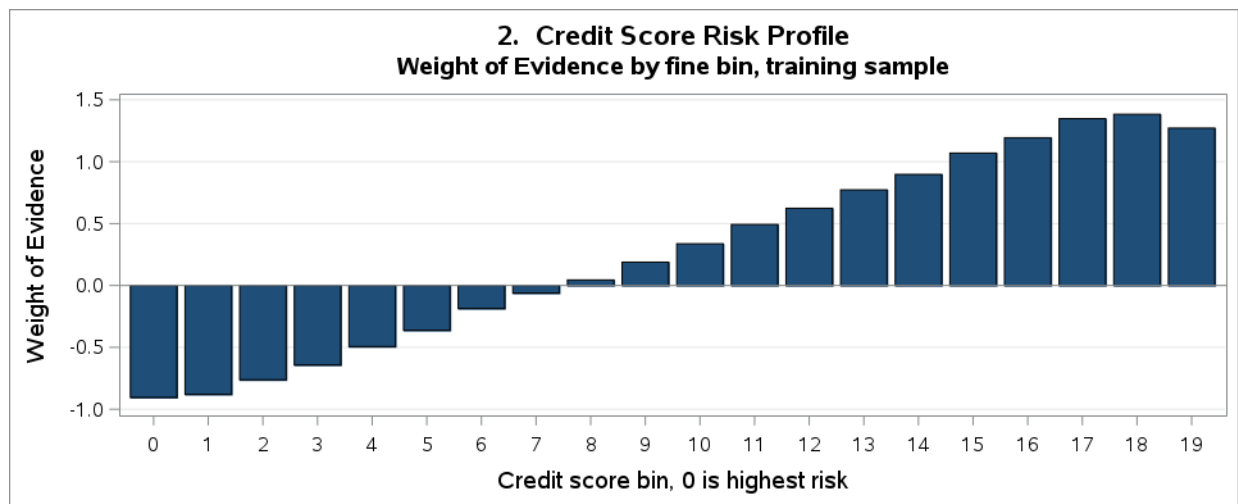
1. Portfolio Composition and Observed Default
Observed default rate by origination year



2. Credit Score Risk Profile

Weight of Evidence by fine bin, training sample

- Each bar is one twentieth of the training population, weakest credit scores on the left, strongest on the right
- Weight of Evidence compares a bin's share of loans that repaid with its share that defaulted
- Bars below zero hold more than their share of defaults; bars above zero hold fewer
- The climb from minus 0.90 to plus 1.38 is steady and almost perfectly ordered
- Information Value of 0.53 on 16.8 million loans, the strongest characteristic tested



Bin	From	To	Loans	Defaults	WOE
0	300	619	822782	38626	-0.9041
1	620	641	851565	39055	-0.8796
2	642	656	817012	33488	-0.7622
3	657	669	855263	31241	-0.6423
4	670	680	831518	26311	-0.4937
5	681	690	804634	22411	-0.3622
6	691	700	883886	20737	-0.1861
7	701	709	802200	16661	-0.0615
8	710	718	811768	15193	0.0447
9	719	727	863097	14027	0.1884
10	728	736	871436	12234	0.3370
11	737	744	810577	9759	0.4927
12	745	752	895382	9470	0.6237
13	753	759	823572	7517	0.7725

2. Credit Score Risk Profile

Weight of Evidence by fine bin, training sample

Bin	From	To	Loans	Defaults	WOE
14	760	765	723979	5838	0.8975
15	766	772	868956	5906	1.0697
16	773	779	864085	5192	1.1937
17	780	786	838108	4320	1.3479
18	787	795	857214	4266	1.3832
19	796	850	814008	4528	1.2713

3. The Scorecard

Points by characteristic and class

- Add the points for each of the three characteristics to obtain the loan score
- Higher points mean lower risk
- 600 points corresponds to 50 to 1 odds of repayment
- Every 20 additional points doubles those odds

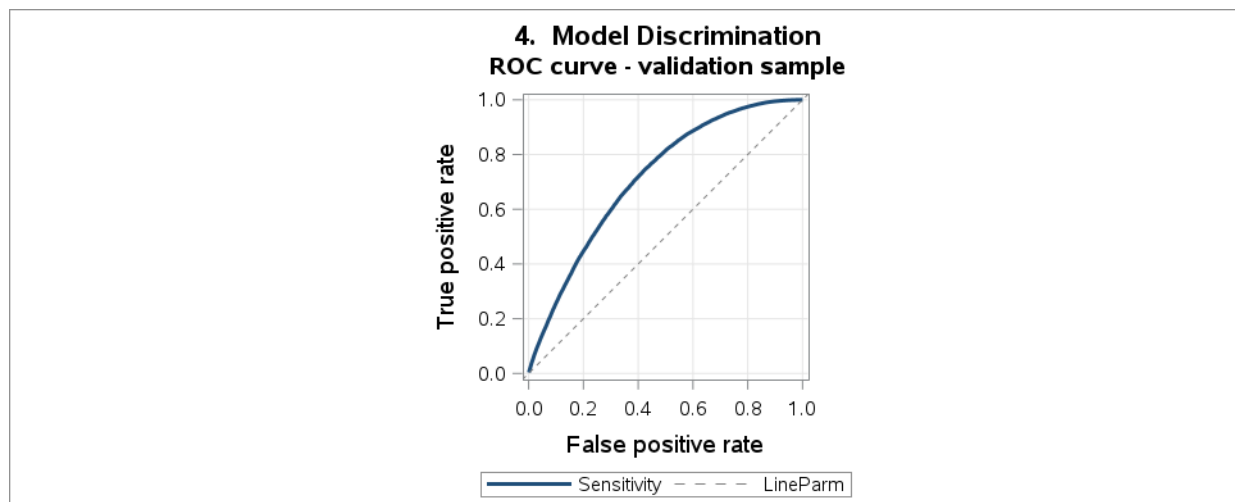
Characteristic	Class	WOE	Points
Credit score	780 and above	1.3341	233
Credit score	766 to 779	1.1297	228
Credit score	753 to 765	0.8291	220
Credit score	737 to 752	0.5593	214
Credit score	719 to 736	0.2604	206
Credit score	701 to 718	-0.0094	200
Credit score	681 to 700	-0.2737	193
Credit score	657 to 680	-0.5716	186
Credit score	620 to 656	-0.8237	180
Credit score	below 620	-0.9041	178
LTV	43 and below	2.0554	248
LTV	44 to 54	1.3212	231
LTV	55 to 62	0.7757	218
LTV	63 to 69	0.3166	207
LTV	70 to 79	-0.0406	199
LTV	exactly 80	-0.2582	194
LTV	81 to 89	-0.5849	186
LTV	90 to 94	-0.7353	183
LTV	95 and above	-0.8508	180
DTI	19 and below	0.6807	211
DTI	20 to 25	0.4392	207
DTI	26 to 32	0.1615	203
DTI	33 to 37	-0.0843	199
DTI	38 to 44	-0.2870	195
DTI	45 and above	-0.3970	194

4. Model Discrimination

Out-of-time validation

- Discrimination asks whether the model puts riskier loans ahead of safer ones
- KS is the widest gap between defaults captured and non-defaults captured, working from worst score to best
- The ROC curve shows the same separation: The further it bows above the diagonal, the better the ranking
- The model was fitted on 2000 to 2005 loans and applied untouched to 2006 and 2007 loans it had never seen
- KS falls only from 37.3 to 32.1, and area under the curve from 0.746 to 0.716
- The ranking held: Riskier loans still scored worse and safer loans still scored better, on loans originated after the training period
- Ranking correctly is not the same as predicting correctly: Section 5 shows the predicted level of default was far too low

Sample	KS
Training	37.26%
Validation	32.05%



5. Calibration - Predicted versus Actual

By score decile

- Discrimination asks whether the ordering is right; Calibration asks whether the predicted level is right
- A model can rank every loan correctly and still be badly wrong about how many will default
- Ordering is flawless: Predicted default rises from 0.19 to 6.28 percent across deciles, actual from 0.54 to 19.64 percent
- Every decile is in the right place, and every decile is understated by a factor of three to four
- 2006 originations: Predicted default rate 2.10 percent, actual default rate 8.17 percent
- 2007 originations: Predicted default rate 2.26 percent, actual default rate 8.94 percent
- The model barely separated the two vintages because the applicants barely differed
- The difference was in the environment those loans matured into, which no origination characteristic can express

Score decile	Loans	Predicted default rate	Actual default rate
0	217,543	0.19%	0.54%
1	226,981	0.49%	2.01%
2	215,601	0.74%	3.56%
3	213,712	0.99%	4.94%
4	218,875	1.34%	6.97%
5	223,765	1.81%	8.65%
6	210,520	2.44%	10.76%
7	230,780	3.24%	12.76%
8	208,570	4.26%	15.75%
9	224,304	6.28%	19.64%

Origination year	Loans	Predicted default rate	Actual default rate
2006	1,079,344	2.10%	8.17%
2007	1,111,307	2.26%	8.94%

6. Population Stability

Training versus validation score distribution

- PSI compares the score distribution of the new population against the population the model was built on
- Below 0.10 is stable and requires no action; 0.10 to 0.25 is a minor shift; Above 0.25 calls for investigation or a rebuild
- The index comes to 0.031, comfortably inside the stable range
- Applicants arriving in 2006 and 2007 looked very much like those the model was trained on
- A monitoring report built on this measure would have raised no alarm while the book defaulted at four times the predicted rate
- The scorecard method worked: It ranked risk correctly and its inputs stayed stable
- What failed was the assumption that the relationship between borrower characteristics and default holds steady over time
- Population monitoring cannot detect that failure because it watches applicants rather than outcomes
- Actual-versus-predicted tracking, shown in Section 5, is what would have caught it, within months of the 2006 vintage seasoning

Score band	Predicted default at band top	Training loans	Validation loans	Percent training	Percent validation	PSI contribution	Population Stability Index
0	0.239%	1,675,873	156,741	9.98%	7.15%	0.0094	.
1	0.475%	1,683,158	167,076	10.02%	7.63%	0.0065	.
2	0.715%	1,702,195	193,490	10.14%	8.83%	0.0018	.
3	0.980%	1,680,025	270,545	10.00%	12.35%	0.0049	.
4	1.367%	1,655,808	204,413	9.86%	9.33%	0.0003	.
5	1.851%	1,703,481	222,848	10.14%	10.17%	0.0000	.
6	2.509%	1,649,701	222,389	9.82%	10.15%	0.0001	.
7	3.341%	1,691,477	230,521	10.07%	10.52%	0.0002	.
8	4.569%	1,632,849	241,656	9.72%	11.03%	0.0017	.
9	9.569%	1,719,381	280,972	10.24%	12.83%	0.0058	0.0308

Appendix. Data Processing Record

ETL Pipeline Processing Record

Sequence	Acquisition quarter	Source file	Monthly records read	Loans after collapse	Defaults identified	Processing time	Cumulative records
1	2000Q1	2000Q1.csv.gz	9,236,094	246,862	3,390	0:00:58.66	9,236,094
2	2000Q2	2000Q2.csv.gz	8,344,168	274,339	3,556	0:00:53.35	17,580,262
3	2000Q3	2000Q3.csv.gz	8,869,382	333,498	3,691	0:00:56.99	26,449,644
4	2000Q4	2000Q4.csv.gz	10,358,630	363,892	4,824	0:01:06.98	36,808,274
5	2001Q1	2001Q1.csv.gz	15,838,528	471,277	6,228	0:01:41.93	52,646,802
6	2001Q2	2001Q2.csv.gz	32,114,758	847,128	9,606	0:03:25.03	84,761,560
7	2001Q3	2001Q3.csv.gz	28,527,792	790,095	9,278	0:03:01.70	113,289,352
8	2001Q4	2001Q4.csv.gz	38,135,841	896,643	10,576	0:04:01.68	151,425,193
9	2002Q1	2002Q1.csv.gz	42,663,862	968,761	11,277	0:04:31.44	194,089,055
10	2002Q2	2002Q2.csv.gz	27,092,919	669,494	9,807	0:02:52.36	221,181,974
11	2002Q3	2002Q3.csv.gz	33,649,174	751,084	10,919	0:03:33.01	254,831,148
12	2002Q4	2002Q4_01.csv.gz	30,000,009	514,243	5,560	0:03:08.52	284,831,157
13	2002Q4	2002Q4_02.csv.gz	30,000,088	514,894	5,558	0:03:10.83	314,831,245
14	2002Q4	2002Q4_03.csv.gz	12,843,706	219,803	2,388	0:01:21.02	327,674,951
15	2003Q1	2003Q1_01.csv.gz	30,000,015	450,411	5,435	0:03:09.84	357,674,966
16	2003Q1	2003Q1_02.csv.gz	30,000,062	451,483	5,512	0:03:10.33	387,675,028
17	2003Q1	2003Q1_03.csv.gz	30,000,001	450,636	5,597	0:03:10.23	417,675,029
18	2003Q1	2003Q1_04.csv.gz	5,011,897	75,328	893	0:00:31.84	422,686,926
19	2003Q2	2003Q2_01.csv.gz	30,000,063	390,956	5,694	0:03:12.34	452,686,989
20	2003Q2	2003Q2_02.csv.gz	30,000,117	389,669	5,692	0:03:09.79	482,687,106
21	2003Q2	2003Q2_03.csv.gz	30,000,052	390,367	5,569	0:03:10.14	512,687,158
22	2003Q2	2003Q2_04.csv.gz	30,000,005	389,778	5,704	0:03:12.16	542,687,163
23	2003Q2	2003Q2_05.csv.gz	7,056,863	92,187	1,340	0:00:45.34	549,744,026
24	2003Q3	2003Q3_01.csv.gz	30,000,068	346,492	5,622	0:03:09.86	579,744,094
25	2003Q3	2003Q3_02.csv.gz	30,000,258	347,173	5,538	0:03:13.74	609,744,352
26	2003Q3	2003Q3_03.csv.gz	30,000,009	346,625	5,650	0:03:10.98	639,744,361
27	2003Q3	2003Q3_04.csv.gz	30,000,072	346,995	5,511	0:03:10.72	669,744,433
28	2003Q3	2003Q3_05.csv.gz	30,000,040	347,202	5,616	0:03:10.02	699,744,473
29	2003Q3	2003Q3_06.csv.gz	363,299	4,125	51	0:00:02.34	700,107,772
30	2003Q4	2003Q4_01.csv.gz	30,000,091	371,069	7,848	0:03:08.24	730,107,863
31	2003Q4	2003Q4_02.csv.gz	30,000,004	370,986	7,892	0:03:08.42	760,107,867
32	2003Q4	2003Q4_03.csv.gz	7,949,259	98,333	2,158	0:00:50.01	768,057,126
33	2004Q1	2004Q1.csv.gz	35,954,755	452,471	12,197	0:03:45.92	804,011,881
34	2004Q2	2004Q2.csv.gz	51,180,070	614,487	16,335	0:06:18.22	855,191,951

Appendix. Data Processing Record

ETL Pipeline Processing Record

Sequence	Acquisition quarter	Source file	Monthly records read	Loans after collapse	Defaults identified	Processing time	Cumulative records
35	2004Q3	2004Q3.csv.gz	28,708,821	389,487	12,398	0:03:29.23	883,900,772
36	2004Q4	2004Q4.csv.gz	28,695,574	361,659	14,160	0:03:30.42	912,596,346
37	2005Q1	2005Q1.csv.gz	24,701,544	303,611	12,921	0:03:00.00	937,297,890
38	2005Q2	2005Q2.csv.gz	27,084,301	339,372	17,269	0:03:18.17	964,382,191
39	2005Q3	2005Q3.csv.gz	35,679,987	440,521	26,096	0:04:24.82	1,000,062,178
40	2005Q4	2005Q4.csv.gz	29,647,091	378,311	25,556	0:03:37.75	1,029,709,269
41	2006Q1	2006Q1.csv.gz	18,179,171	253,043	19,135	0:02:12.45	1,047,888,440
42	2006Q2	2006Q2.csv.gz	19,718,207	291,165	23,010	0:02:24.65	1,067,606,647
43	2006Q3	2006Q3.csv.gz	16,542,961	271,373	21,450	0:02:01.52	1,084,149,608
44	2006Q4	2006Q4.csv.gz	18,044,494	280,979	23,835	0:02:11.97	1,102,194,102
45	2007Q1	2007Q1.csv.gz	16,759,833	253,279	21,612	0:02:03.54	1,118,953,935
46	2007Q2	2007Q2.csv.gz	18,587,946	287,263	24,650	0:02:16.33	1,137,541,881
47	2007Q3	2007Q3.csv.gz	18,293,962	314,703	26,392	0:02:14.94	1,155,835,843
48	2007Q4	2007Q4.csv.gz	22,672,192	391,181	38,512	0:02:47.15	1,178,508,035

Data sets processed	Total monthly records	Total loans	Total defaults	Total SAS processing time	Records per second
48	1,178,508,035	19,144,733	519,508	2:10:57	149,997

